

HOOO EP 2026 Standard

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In this paper I present a 2026 standard for HOOO EP. This standard will be used as part of Standard Path Semantics in the future, until a new standard is published. The 2026 Standard weakens the 3rd axiom in HOOO EP that measures meta-strength using a theorem grading, and introduces an additional theorem grading for collapse-strength.

HOOO EP stands for “Higher Order Operator Overloading Exponential Propositions” and is the basis for unifying meta- and object-language in IPL (Intuitionistic Propositional Logic) using the Curry-Howard correspondence of function pointer types in Logic. It is used in Path Semantics to reason about operators that are tautological congruent, but not normal congruent. This theory is needed to study the logical foundation of Path Semantics, particularly the core axiom, where the path semantical quality and path semantical qubit operators are tautological congruent.

The HOOO EP 2026 Standard is specified as the two following theorem gradings:

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grade meta_strength [pow_lift, tauto_hooo_imply, tauto_hooo_imply_or];
grade collapse_strength [tauto_hooo_tauto_or, tauto_hooo_or];
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Where:

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pow_lift : a^b -> (a^b)^c
tauto_hooo_imply : (a => b)^c -> (a^c => b^c)^true
tauto_hooo_imply_or : (a | b)^c -> ((c => a) | (c => b))^true
tauto_hooo_tauto_or : (a | b)^true -> (a^true | b^true)^true
tauto_hooo_or : (a | b)^c -> (a^c | b^c)^true
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At both meta-strength 3 and collapse-strength 2, IPL becomes stronger than EL (Existential Logic, also known as De Morgan Logic or Weak Excluded Middle). However, meta-strength 3 is safe from collapsing paradoxes with negative propositions, unlike collapse-strength 2.

EL can be thought of as a type theory where every type is either inhabited or uninhabited. The idea of high meta-strength or collapse-strength, is that for every sum type, there is a map from the context where the sum type is constructed to some sum type variant. In Set Theory, this is similar to the Axiom of Choice.

At collapse-strength 1, plus a proposition in EL, paradoxes collapse with negative propositions for that proposition. Similarly, at collapse-strength 1, plus a proposition in PL (Propositional Logical with Excluded Middle), both tautologies and paradoxes collapse with respective normal propositions. Even though collapse-strength 1 itself neither implies EL or PL, it is not safe to use with these logics, if one wants to preserve the notion of possible world semantics in theorem proving, e.g. necessity, contingency and impossibility as 3 separate modalities (interpreted in HOOO EP, not in Modal Logic). When tautologies and paradoxes collapse with respective normal propositions, the semantics becomes only a single possible world, which corresponds to pure PL.

High collapse-strengths are still interesting to study, despite their collapsing properties, due to the Time Interpretation of the core axiom in Path Semantics. Under this interpretation, moments in time behave approximately to Newtonian universal time, where each moment, described by a path semantical layer, can be thought of a space for normal reasoning in Logic. Here, collapse is interesting to study, because observer moments are still unknown relative to Everett branches.

The enormous complexity of an Everett many-worlds interpretation of quantum mechanics makes it nearly impossible for scientists to study what makes observer moments possible, which is made even more complex by the possibility of an extreme observer selection effect, due to cosmic Eternal Inflation, predicted by most theoretical models of the early universe. A proposal in Path Semantics is to modify Wolfram models, which are based on hypergraph rewriting systems, to include matching against distinct nodes, as avatar-hypergraph rewriting systems. This might help to explain why the universe is seemingly complex, instead of a very simple universe where the rewriting rule matches all the time. The idea is that the amount of observer moments are correlated with events where the rewriting rule matches against distinct nodes, something that is impossible to express in normal constructive logic. Using the perspective of the core axiom of Path Semantics, this might be thought of as the lifting axiom from propositional equality and symbolic distinction into path semantical quality. This means, since this HOOO EP 2026 Standard allows studying meta-strength separated collapse-strength, might open up a path forward for better understanding observers.

Another research idea in Path Semantics is that physical reality has a particular mode of evaluation which preserves unitary properties, the ability to reverse time in principle. In theoretical physics this is related to the Holographic Principle, developed to study events close to black holes in space-time. In Path Semantics, this is called a Physical Seshatic Identity and can be thought of as a way to trace metaphysical identity of a single bit of information through a “holomaze”. In this puzzle, it is impossible for the player to move in such a way that their path backward in time is undetermined.

The Time Interpretation of the core axiom in Path Semantics allows multiple histories of observer moments. Therefore, there might be alternative versions of the core axiom out there to study, where an observer moment only has one possible past. There have been proposals of multiple versions of the core axiom, but none of these versions have been studied in much detail. This means, the apparent collapse of the wave function from the perspective of the observer, which might be due to decoherence or something else, might have a source in the particular version of the core axiom being used, or perhaps meta-strength and collapse-strength also play a role. We do not know yet, because basically nothing about Path Semantics has been studied to a great detail so far.

The basic idea here is that there is good reason to treat meta-strength and collapse-strength as separate theorem gradings, motivated by potential future research on theories of observer moments.